

Re-examining the mortality contrast between high and low central adiposity in non-obese MASLD

Model submission — M4 reanalysis letter. Group A. Correspondence on Kueh et al., BMJ Open 2026;16:e113719. All figures are regenerated by the repository accompanying this letter; the locked-contrast values typed here are checked at build time against `_results/chain.json`, which the analysis emits.**

Kueh and colleagues report that among adults with MASLD, non-obesity with high central adiposity confers the poorest survival, and that this holds independent of total body weight. We reproduced their analysis from the public NHANES 2007–2018 files. The replication comes close, though it does not match exactly. Our disagreement is not with their arithmetic but with which contrast supports the stated conclusion, and with how far that conclusion travels.

We set out what stands, what we would re-price, what we think should be corrected, and what none of our analyses resolve.

What stands

The numerical replication. Rebuilding the cohort from raw microdata comes close to the published figures without matching them exactly. Our analytic file has 6,371 rows against the paper’s approximately 6,300, and we have not been able to explain the difference: the alcohol exclusion the Methods describe does not account for it. Group sizes and deaths are close (586 deaths against 585). Under the paper’s Model 1, unweighted, on the rows of our full analytic file complete on its covariates, Group IV against Group I gives 3.29 against the published 3.155. Each difference is logged in the accompanying repository, and those we could not explain are marked as unresolved. We found no error in the authors’ computation.

Robustness to incomplete covariate data. Cancer status is the only Model 1 covariate carrying missingness. On the contrast we defend (Group IV vs III, design-weighted), and holding the authors’ covariate set exactly, complete-case analysis on the 5,939 rows complete on Model 1 gives 1.29 (0.89–1.86). Both extreme assignments, and multiple imputation with the outcome in the imputation model, keep all 6,048 rows and give 1.28 (0.89–1.85). A difference of 0.01 is far smaller than the interval: the estimate does not depend in any meaningful way on how the incomplete records are handled.

We stress the scope of that finding, because it is narrower than it may appear. It concerns people **already in the analytic cohort**. It is evidence against one specific alternative explanation — that the result is an artefact of dropping incomplete records — and it is not evidence that the estimate would hold in anyone excluded before cohort entry. Robustness is not transportability, and we do not offer the first as an answer to questions about the second.

What we would re-price

The comparison that supports the headline claim. The paper’s conclusion is comparative: it says one phenotype fares worst. That is a statement about Group IV against Group III. The reported models are estimated against Group I.

Read crudely, the ranking holds and the paper reports it accurately. Read under the paper’s Model 1, the picture changes. The published Model 1 gives Group III 3.175 and Group IV 3.155 against Group I — an implied IV-versus-III ratio of about 0.99. Estimating that contrast directly, design-aware, on the authors’ own Model 1 covariate set and the 5,939 complete-case rows, we obtain a hazard ratio of 1.29 (0.89–1.86), compatible with no difference as well as with a meaningful increase.

We report that estimate here as our defended result, and we note that the published table is *consistent* with it. We are careful not to claim more: the implied ratio cannot be given an interval, because the covariance of the two published coefficients is not reported. This is agreement, not independent corroboration — a distinction our own earlier draft blurred, and one our reviewers were right to press.

How much weight the estimand problem should carry. At M1 we named estimand–adjustment-set alignment as the single dominant threat: the target quantity Model 1 represents is never stated, so it cannot be established that the adjustment set is the right one, and the attenuation has no unique interpretation. We continue to think this is the prior question. But we now think it is prior rather than larger — it determines whether there is an interpretable estimate at all, while the limitation in the next section determines where that estimate applies. Ranking them against each other was less useful than separating them.

What should be corrected

The population the conclusion is addressed to. The analytic cohort is not simply a sample of US adults with MASLD. It is reachable only through two successive restrictions, and they are different in kind:

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adults 18+	36,580
sampled into the fasting subsample	14,962
of those, with a computable Fatty Liver Index	14,170

The first is **selection by design**. NHANES chose who was asked to fast, and WTSAF2YR exists to account for it. Weighting by it, and subsetting the survey design rather than the data, carries the estimate from the fasting subsample back to the US adult population it was drawn to represent. This step is handled.

The second is **item-level availability**. Among people already sampled to fast, the index still requires triglycerides, GGT, BMI and waist circumference simultaneously. No design weight accounts for that.

The conclusion should therefore be stated for the population in which the exposure can be constructed, rather than for adults with MASLD generally. This is a narrowing of scope, not a retraction: within that population the descriptive finding stands.

What remains unresolved

Selection at the second step is not repaired by anything we did. Weighting does not address it, because the weights were not built for it. Our missing-data analyses do not address it, because they operate inside the cohort. We cannot determine from the public files why an individual lacks an index component, and therefore cannot establish whether that absence is independent of mortality risk.

If it is not, the contrast itself — not merely its generalisability — is affected. We regard this as the most consequential open question in the analysis, and we state it as unresolved rather than bounded.

How this analysis reached its final form

We record the sequence, because two of the changes came from review and one did not.

At **M1** we replicated the paper, held the replication and scientific contrasts apart, and named estimand–adjustment-set alignment as the dominant threat. At **M2** our reviewers accepted the replication, verified that the survey design was subsetting correctly, declined to re-open the missing-data question we had already answered, and argued that selection into the cohort mattered more than the estimand problem. At **M3** we adopted their framing of selection but decomposed it into the two mechanisms above, which we regard as the single most useful change in the project; we retained our ranking on a narrower argument; and — on our own initiative, unprompted by any reviewer — we ran the missing-data sensitivity on the locked contrast, having deleted an earlier version that widened the adjustment set so the comparison would show movement. That widening would have answered a question nobody asked.

Two of those changes were reviewer-prompted, one was self-initiated, and one reviewer argument we tested and did not accept. We record which was which because responding to review is not the same as complying with it, and a reader is entitled to know which decisions were ours.

Conclusion

The authors' analysis reproduces closely, if not exactly, and their descriptive finding is accurate. We suggest three revisions: state the target quantity the adjusted models are meant to represent; report the Group IV versus Group III contrast directly, since that is the comparison the conclusion makes; and state the conclusion for the population in which the Fatty Liver Index can be computed. The first two are repairable in the text. The third is a limitation to be carried, not solved.

Cover note

Contribution. AB: cohort reconstruction and survey-design code, the two-step selection decomposition. CD: estimand argument and threat re-ranking, letter drafting. EF: missing-data sensitivity on the locked contrast, repository assembly and the fresh-machine run.

External feedback. Received and it changed the submission — see *How this analysis reached its final form*, above. No individual outside the group edited text, wrote material, altered code, ran an analysis, or contributed a figure or table.

AI Use Statement. Tool and version: Claude (Opus 5), in RStudio and in a browser. Where we used it: the imputation-and-refit wrapper in `P5_code.qmd`, and prose editing of the Conclusion. What we did with the output: every generated block was run, checked against the increment it came from, and corrected where it was wrong — the AI-drafted `mice()` call was left unseeded, so two runs of the imputation disagreed in the third decimal, and the fix is the `seed=604` that file now carries. Nothing on the numbers in this letter is offered as the AI's responsibility.

This text also ships as `COVER-NOTE.md` in the root of the submitted repository zip.